

4.6 Spring Boot Actuator

• What is Spring Boot Actuator ?

- Spring Boot Actuator is a **production monitoring** and **management** module that exposes operational **information** about a **running application** through **predefined endpoints**.
- It provides features such as **health checks**, **metrics**, **environment information**, **bean inspection**, and **application monitoring**.
- In production environments, Actuator is commonly integrated with **Micrometer**, **Prometheus**, and **Grafana** to monitor system **health**, **performance**, and **resource utilization**.
- It helps developers and operations teams quickly diagnose issues without directly accessing application servers.

• Installing Spring Boot Actuator

- We need to add this particular dependency in pom.XML, then refresh and let maven to download it's required stuffs.

```
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-actuator</artifactId>
</dependency>
```

• Actuator Config :

- To change the base path of actuator :

```
management.endpoints.web.base-path=/management
```

- Enable or Disable actuator :

```
management.endpoints.web.exposure.include=*
Or
management.endpoints.web.exposure.include=health,info,beans,env
```

- Control authorization:

```
management.endpoint.health.show-details=never
management.endpoint.health.show-details=always
management.endpoint.health.show-details=when-authorized
```

• Actuator Endpoints :

1. **beans**: this endpoint returns the list of all the beans configured in our application.
2. **env**: provides information about the Spring Environment properties.
3. **health**: Shows application health.
4. **info**: Displays application information, we can configure this in Spring environment properties.
5. **mappings**: Displays the list of all *@RequestMapping* paths.
6. **threaddump**: provides the thread dump of the application.

4.6 Spring Boot Actuator

• What are Metrics in Spring Boot Actuator ? :

- Metrics are numerical **measurements** that **describe** the **runtime behaviour** and **health** of an application.
- They help answer questions like:
 - How much **memory** is being used?
 - How many **requests** are coming in?
 - How many **database connections** are **active**?
 - What is the **CPU usage**?
 - How many **threads** are **running**?

• How can we Access Metrics :

- Here we can see all the metrics.

Request	Response
GET /actuator/metrics	<pre>{ "names": ["jvm.memory.used", "jvm.gc.pause", "http.server.requests", "hikaricp.connections.active", "system.cpu.usage"] }</pre>

- For specific metric.

Request	Response
GET /actuator/metrics/jvm.memory.used	<pre>{ "name": "jvm.memory.used", "measurements": [{ "statistic": "VALUE", "value": 1.245E8 }] }</pre>

- Current JVM Memory Used = ~124 MB

• What is /actuator/info ? :

- The **/actuator/info** endpoint is used to expose **custom** application **metadata** such as application **name**, **version**, **build information**, **Git commit details**, **deployment information**, and other business-relevant details.
- It helps **developers** and **operations teams** identify which version of an application is currently running in different environments **without accessing** the **server** directly.

4.6 Spring Boot Actuator

• How to Add Information ? :

- We can add using application.properties:

```
management.info.env.enabled=true
info.app.name=College Management System
info.app.version=1.0.0
info.app.description=Student and Subject Management Platform
info.app.owner=Sibasundar Jena
```

• How to Add Git Information ? :

- We need to add some properties and a plugin in pom.xml:

properties :: Application.properties	plugin :: pom.xml
management.info.git.enabled=true	<pre><plugin> <groupId>io.github.git-commit-id</groupId> <artifactId>git-commit-id-maven-plugin</artifactId> <version>9.0.1</version> <executions> <execution> <goals> <goal>revision</goal> </goals> </execution> </executions> </plugin></pre>

- After adding this clean maven (mvn clean) and again re-compile (mvn package) it.
- Outcome is noticeable:

```
{
  "app": {
    "name": "College Management System",
    "version": "1.0.0",
    "description": "Student, Subject and Faculty Management Platform",
    "owner": "Sibasundar Jena"
  },
  "git": {
    "branch": "main",
    "commit": {
      "id": "c1e6dcd",
      "time": "2026-06-04T16:48:37Z"
    }
  }
}
```

- We can add build, java, OS informations by adding just some properties and get the info by actuators.

```
management.info.build.enabled=true
management.info.java.enabled=true
management.info.os.enabled=true
```